

C Programming

Overview

C is a procedural programming language that provides direct control over memory and hardware resources. It has influenced many later languages and remains important for operating systems, embedded systems, and performance-sensitive software.

Variables and Memory

C uses statically declared types such as `int`, `float`, `double`, and `char`. Variables occupy memory, and programmers can use pointers to store and manipulate memory addresses.

Pointers

Pointers are one of C's defining features. A pointer can refer to another object's memory address, allowing functions and data structures to work directly with memory.

Functions and Arrays

C programs are organized into functions. Arrays store elements of the same type in contiguous memory, and strings are commonly represented as arrays of characters terminated by a null character.

Applications

C is widely used for operating systems, embedded systems, device drivers, compilers, networking software, and other applications where predictable performance and low-level control are important.